SYSTEM FUNDAMENTALS COURSE · MODULE 1

Introduction to Computer Systems

Hardware, software, architecture, and the factors that shape performance

Computer System Components · Hardware & Software · Performance Factors

WHAT WE'LL COVER

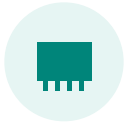
Learning Topic



01
Hardware vs. Software



02
The CPU



03
Memory



04
Storage



05
Input / Output Devices



06
The Operating System



07
Hardware–Software Interaction

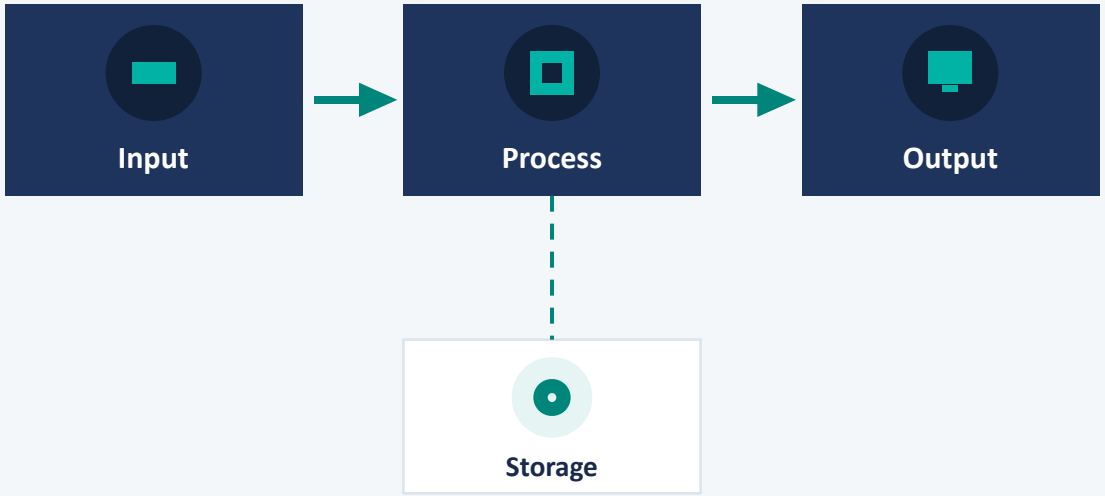


08
Factors Affecting Performance

What Is a Computer System?

A computer system is an organized combination of physical components and instructions that work together to accept data, process it, and produce useful results.

- Hardware supplies the physical machinery
- Software supplies the instructions
- Together they turn raw data into information



THE TWO HALVES OF EVERY SYSTEM

Hardware vs. Software

Hardware

The physical, tangible parts of a computer you can touch.

- Processor (CPU) and circuit boards
- Memory chips (RAM/ROM)
- Storage drives (SSD/HDD)
- Input & output devices

Software

The instructions and data that tell hardware what to do.

- Operating systems
- Application programs
- Drivers connecting hardware
- Data and configuration files

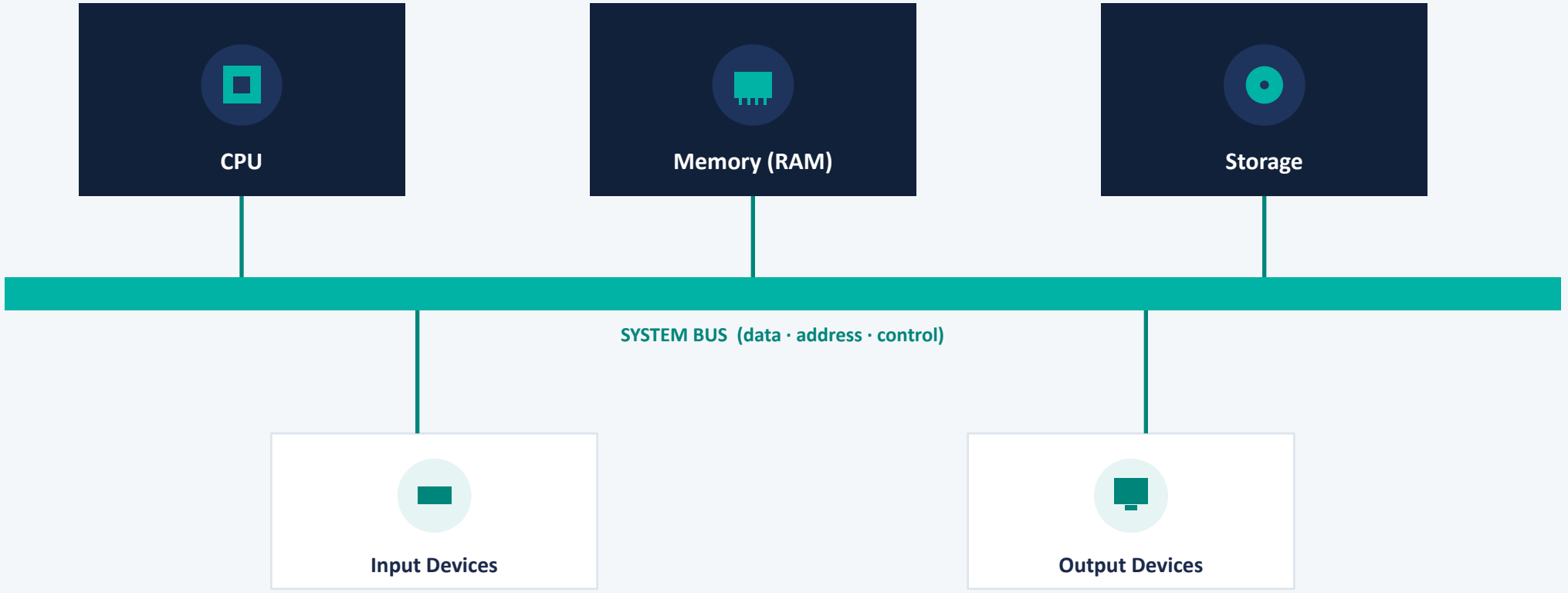
COMPARING THE TWO

Key Differences

Aspect	Hardware	Software
Nature	Physical, tangible	Logical, intangible
Can you touch it?	Yes	No
Wears out?	Physically over time	Doesn't wear, but can become outdated or buggy
Changed by	Replacing or upgrading parts	Updating, patching, or reinstalling
Examples	CPU, RAM, keyboard, monitor	Windows, a web browser, a game

Basic Computer Architecture

Most computers follow the von Neumann model: a CPU, memory, and I/O devices connected by a shared system bus that carries data, addresses, and control signals.



02

The CPU

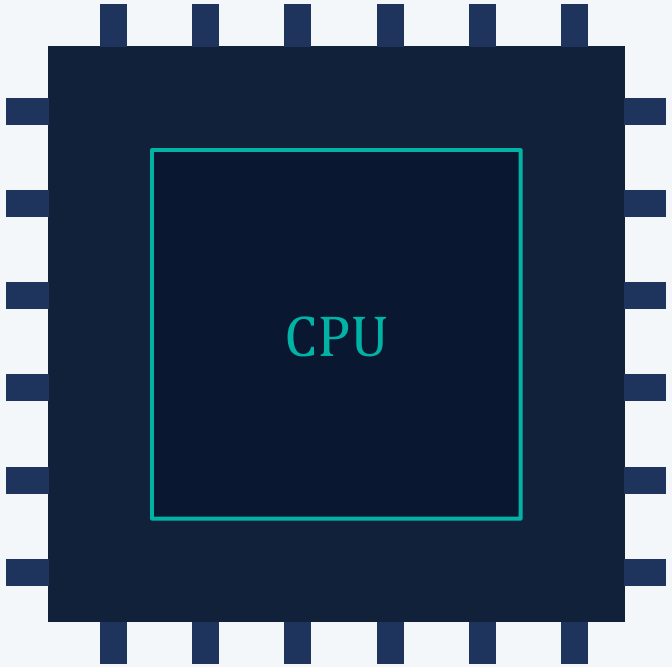
The processor — where instructions are executed



What Is the CPU?

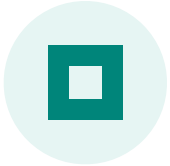
The CPU is the primary component that carries out instructions from programs by performing arithmetic, logic, control, and input/output operations. It's often called the “brain” of the computer.

- **Speed:** Measured in GHz — cycles executed per second. GHz tells you the clock rate, not the complete performance of the CPU.
- **Cores:** individual processing unit inside a CPU
- **Role:** Executes every instruction a program issues



UNDER THE HOOD

Inside the CPU: Three Core Parts



ALU

Arithmetic Logic Unit

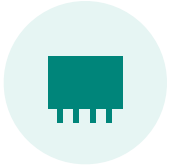
Performs mathematical calculations and logical comparisons (>, <, =).



CU

Control Unit

Directs and coordinates operations — fetching and interpreting instructions.

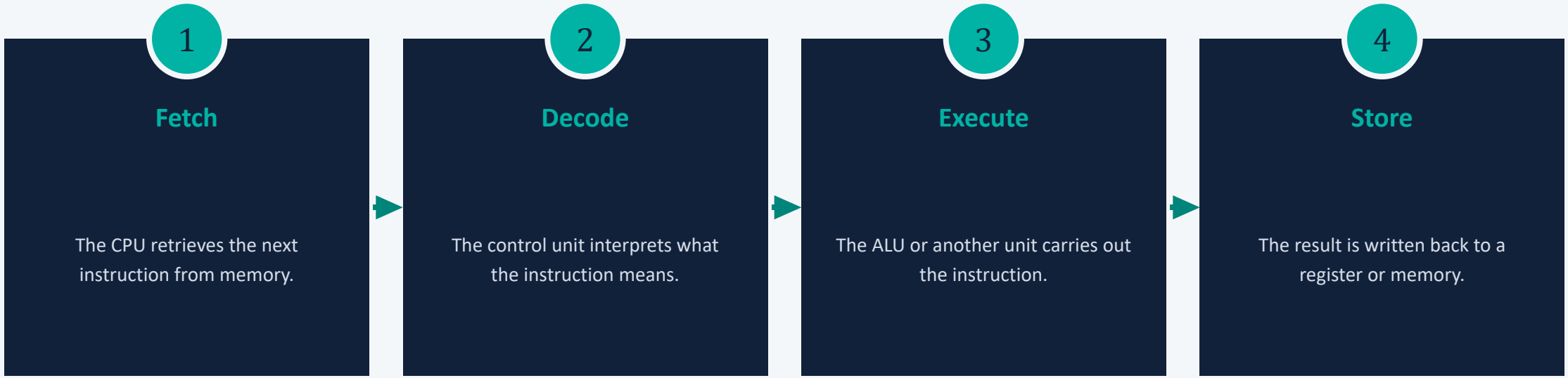


Registers

Registers

Small, ultra-fast storage inside the CPU holding data being worked on right now.

The Fetch–Decode–Execute Cycle



This cycle repeats billions of times per second — the faster it runs, the faster the CPU processes instructions.

CPU Performance: Clock Speed & Cores

GHz

Clock Speed

Number of cycles the CPU completes each second — higher generally means faster processing.

Cores

Multiple Cores

Each core can process its own instructions, allowing several tasks to run in parallel.

Cache

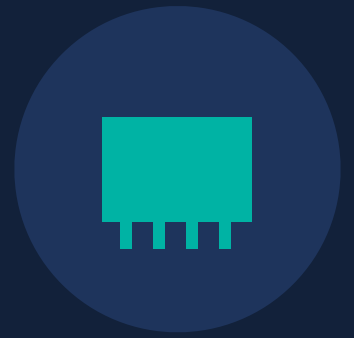
CPU Cache

A small pool of fast on-chip memory that reduces waiting time for frequently used data.

03

Memory

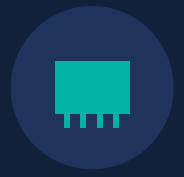
The system's short-term, high-speed workspace



SHORT-TERM WORKSPACE

What Is Memory?

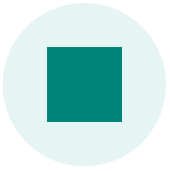
Memory temporarily holds data and instructions that the CPU needs right now. It is much faster than storage but loses data differently, depending on its type.



RAM

Random Access Memory

Holds actively running programs and data. Volatile — contents clear when power is lost.



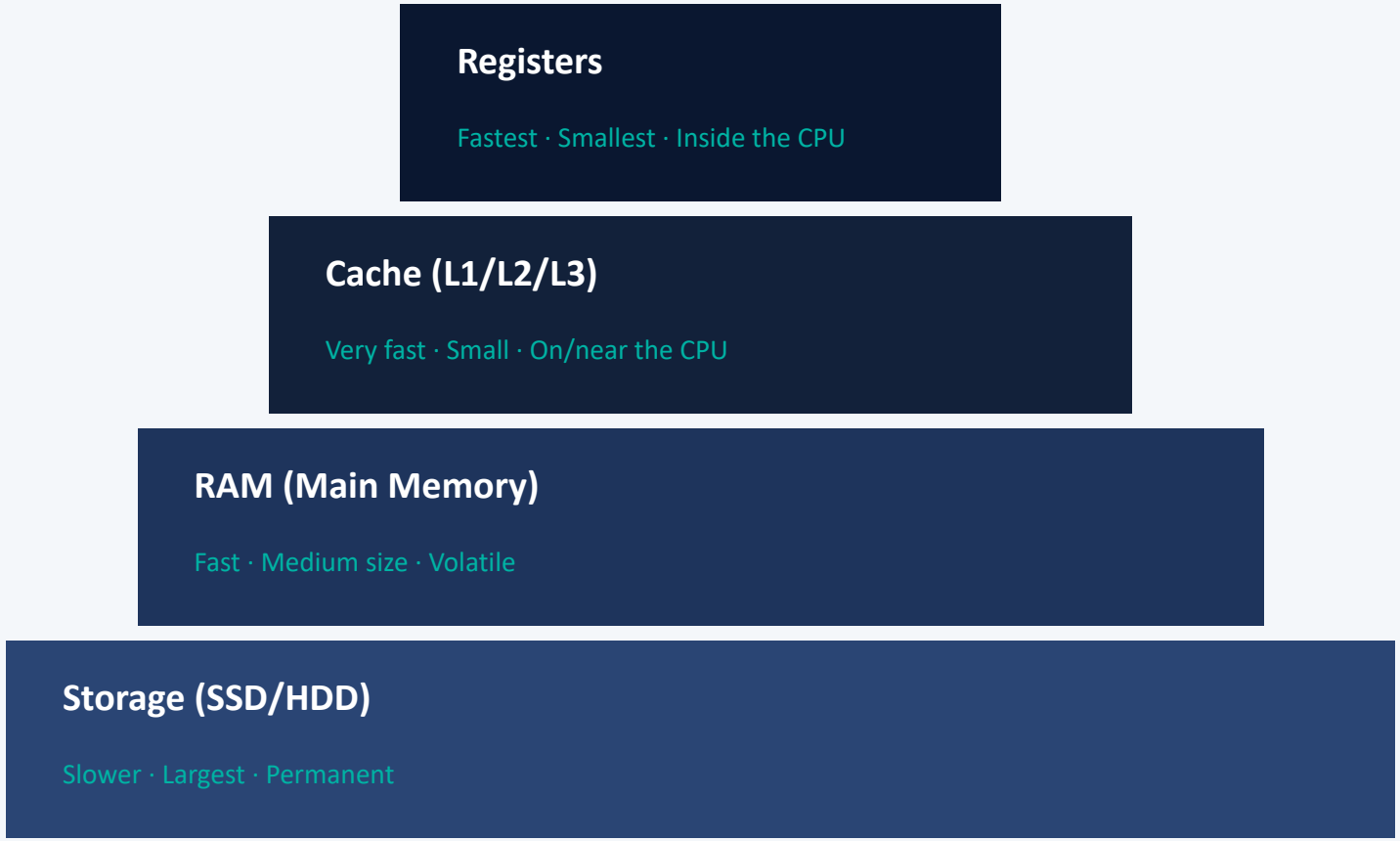
ROM

Read-Only Memory

Stores essential startup instructions (firmware). Non-volatile — retains data without power.

FAST AND SMALL → SLOW AND LARGE

The Memory Hierarchy



TWO KINDS OF MEMORY

RAM vs. ROM

Aspect	RAM	ROM
Volatility	Volatile — cleared when powered off	Non-volatile — retains data without power
Purpose	Temporary workspace for running programs	Stores fixed startup/firmware instructions
Writable?	Read and write, constantly	Typically read-only or rarely rewritten
Speed	Very fast access	Fast, but not usually rewritten
Typical size	8–64 GB in modern PCs	A few MB — just enough for firmware

04

Storage

Where data lives permanently, even without power



LONG-TERM DATA

What Is Storage?

Storage holds data and programs permanently, even when the computer is switched off — unlike memory, which is temporary.



Primary Storage

Built in and fast to access; holds the operating system, installed apps, and everyday files.

EXAMPLES

Internal SSD/HDD



Secondary Storage

Removable or external; used for backups, transfers, and extra capacity.

EXAMPLES

USB drives, external HDDs, optical discs, cloud storage

Storage Types: HDD vs. SSD

Aspect	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Technology	Spinning magnetic disks + moving read/write head	Flash memory chips, no moving parts
Speed	Slower read/write speeds	Much faster read/write speeds
Durability	More vulnerable to physical shock	More durable, no mechanical parts
Noise & heat	Produces noise and more heat	Silent, runs cooler
Cost per GB	Cheaper per gigabyte	More expensive per gigabyte

05

Input & Output

How a computer talks to the outside world



Input Devices

Input devices let users and the environment send data and commands into the computer for processing.



Keyboard

Text and command entry



Mouse / Trackpad

Pointing and selection



Microphone

Audio and voice input

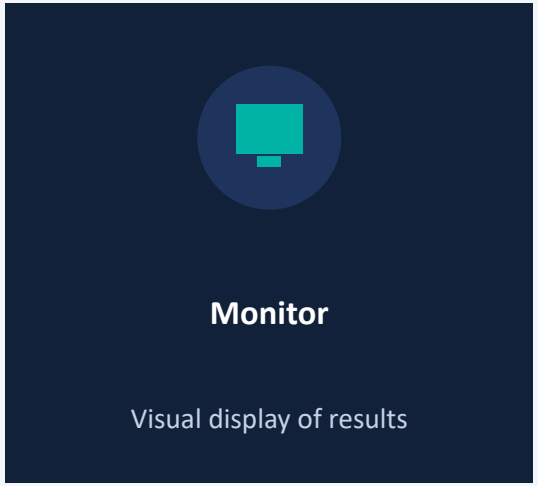


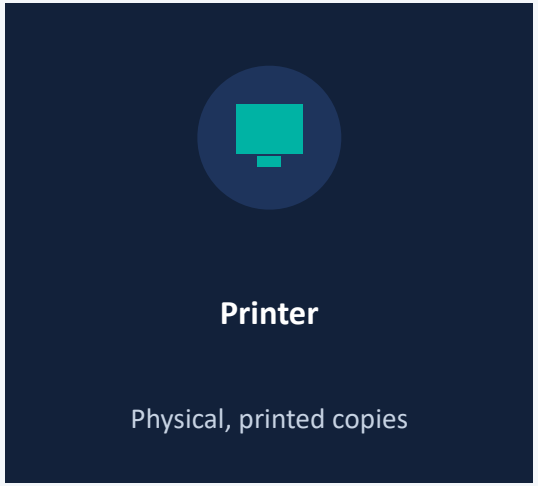
Scanner / Camera

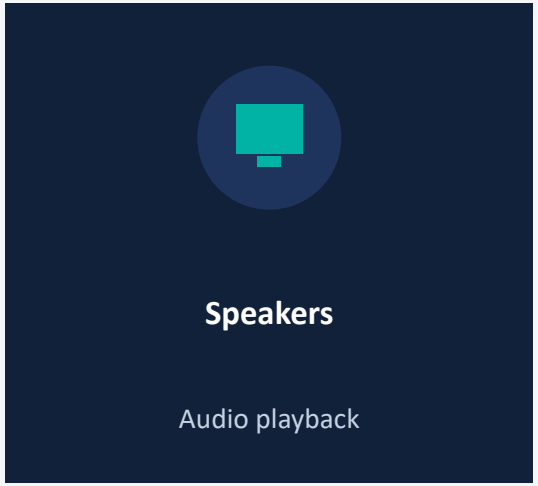
Images and documents

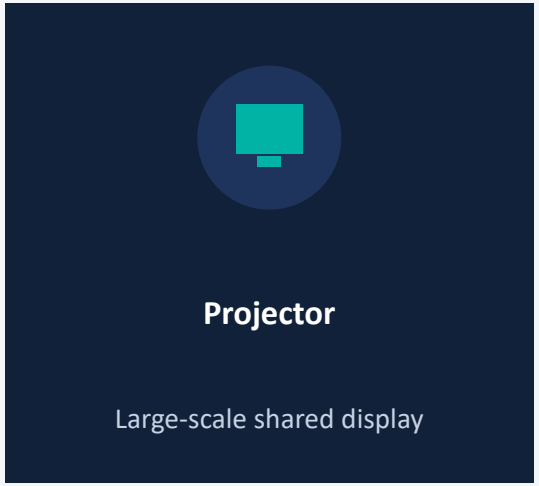
Output Devices

Output devices present processed information back to the user in a usable form.



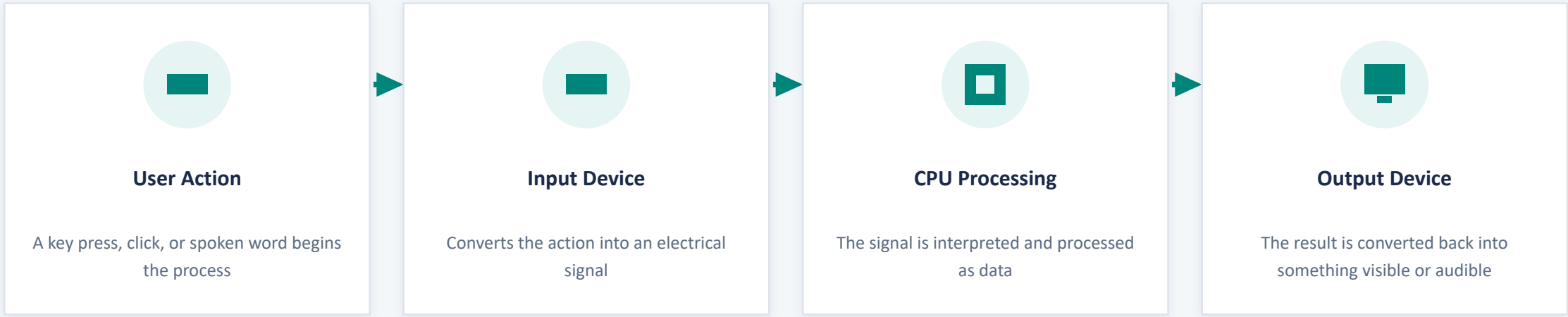






PUTTING IT TOGETHER

How Data Flows Through I/O



06

The Operating System

The software layer that manages everything else

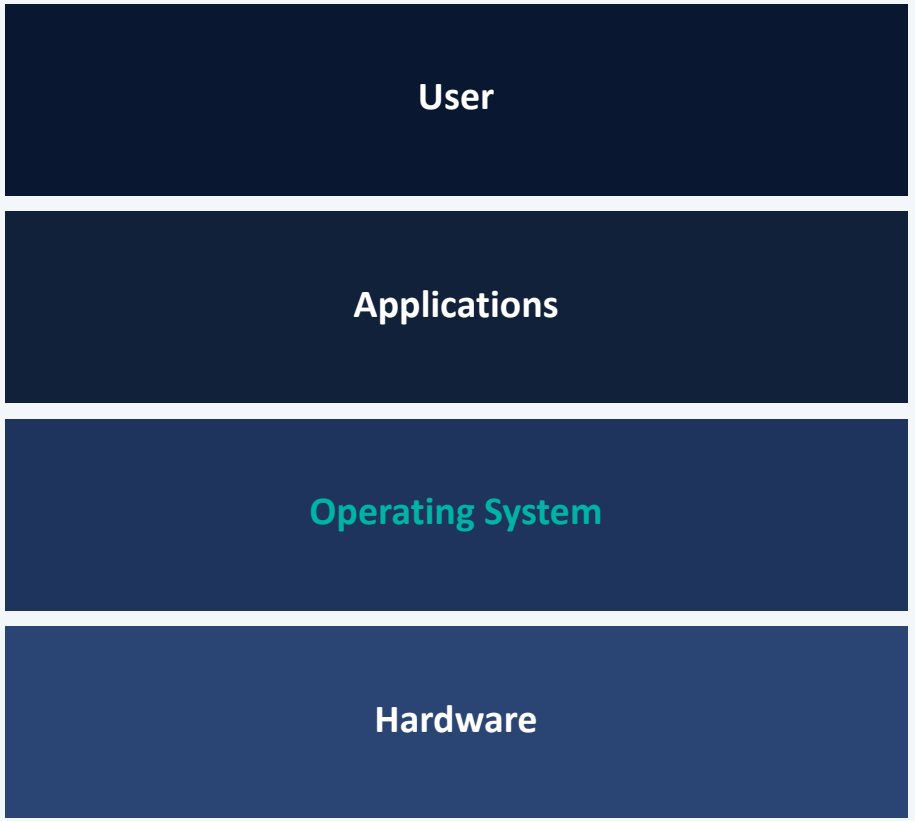


SOFTWARE THAT MANAGES HARDWARE

What Is an Operating System?

An operating system (OS) is system software that manages computer hardware, software resources, and provides common services for application programs. Examples include Windows, macOS, and Linux.

- Acts as an intermediary between users and hardware
- Runs and manages all other software
- Allocates hardware resources fairly and efficiently



Key Operating System Functions

Process Management

Schedules and switches between running programs

Memory Management

Allocates RAM to programs and reclaims it when done

File System Management

Organizes, stores, and retrieves files on storage

Device Management

Controls communication with hardware via drivers

Security & Access

Manages user accounts, permissions, and protection

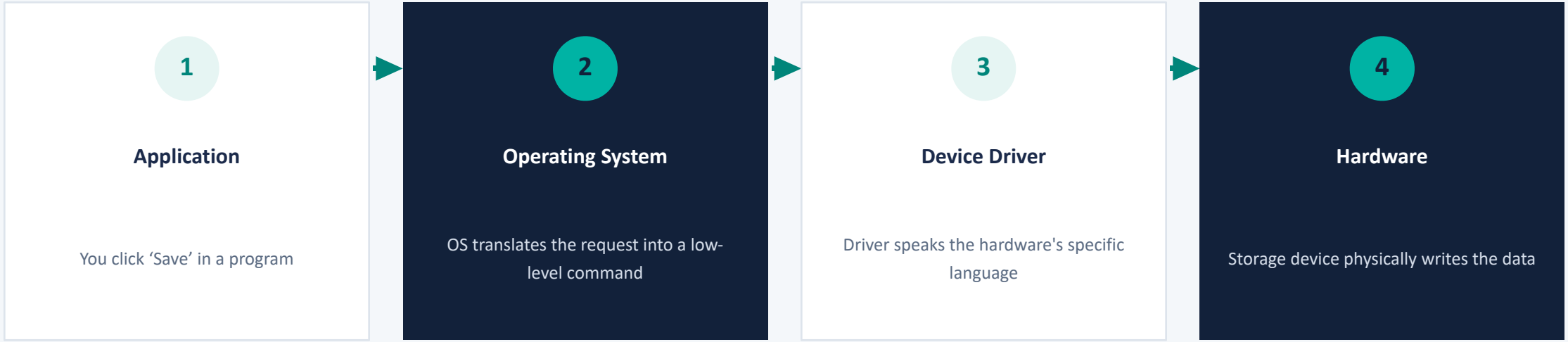
User Interface

Provides the desktop, windows, or command line

TWO HALVES, ONE SYSTEM

How Hardware and Software Interact

Software gives instructions; hardware carries them out. The operating system sits between the two, translating requests from applications into actions the hardware understands.



Factors That Affect Computer Performance



CPU speed & core count

More GHz and cores handle more instructions in parallel



RAM capacity

More memory lets more programs run smoothly at once



Storage type

SSDs load programs and files far faster than HDDs



Software efficiency

Well-optimized code and a lean OS reduce wasted work



Background processes

Too many running tasks compete for the same resources



System heat & age

Overheating and aging components can throttle performance

WRAPPING UP

Key Takeaways

1. Hardware is physical; software is the instructions that drive it
2. The CPU, memory, and storage each play a distinct, specialized role
3. Input and output devices connect the system to the real world
4. The operating system manages hardware and coordinates every program
5. Performance depends on the balance of CPU, memory, storage, and software

Thank You

Questions? Let's discuss.

